



University
of Regina

Go far, *together.*

ENSE 375 – Software Testing and Validation

Group Leader

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- Proof read the text for typing and grammar mistakes.

- Follow the IEEE Bibliography style for the references by selecting "References/ Citations & Bibliography/ Style".

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1 Introduction

- Give a brief description of the design and a summary of the relevant background information related to the topic. Give a rationale about what is needed and why.
- Give the reader an overview of what is in the next sections.
- Do not put any detailed results of your work here.

2 Design Problem

This section has the following two subsections:

2.1 Problem Definition

For this design project, our group defined the problem to be the process of planning and designing, then creating and thoroughly testing a Model View Controller website using proper industry standard testing practices covered in class. Constraints also apply, which we must consider and follow several of them; these include: Economic factors, Regulatory compliance, Reliability, Sustainability and Environmental Factors, Ethics, and Societal Impacts. Due to the nature of the project, the focus of the project is to gain practical experience using several different testing and validation techniques, including Boundary value testing, Equivalence class testing, Decision tables testing, State transition testing, and Use case testing. As well, two functions must be tested; using Path testing and Data flow. During this process, we will follow the structured engineering design process; such as considering requirements, research, and brainstorming. Including analyzing and iterating on potential functions and prototypes. One important idea we are considering is whether to have accounts, which has a large upside, and some downside to implementing, and our process will prototype and test it. Our approved topic/idea for this project is Group Leader, a scheduling app for groups. Each member adds their schedule and availability, then the program will find common times between them, allowing the group to all pick a preferred time. This has a lot of use cases, including just for friend groups to meet, or even in workplaces, when they would need to schedule a meeting. Our first objective is societal impacts, which is to create this website, which will allow users to easily schedule times for meetings between busy schedules. It should be intuitive, quick, and user should have an easy time inputting their schedules. We also aim to meet regulatory compliance and user security risks, by either finding a solution for a secure log in, or avoiding storing schedule information. For reliability, we aim to take advantage of the testing cycles to ensure reliable and stable user experience. For economic factors, we are hoping to complete the project with spending no money, and if databases are required we will try to host locally, the product will also be free. Delegating work is another area for this project, where we will track which group member contributes individually. During the development cycle, we will document this process thoroughly, including test cases we use, commenting code, and having a public github.

2.2 Design Requirements

This section has the following three subsections:

2.2.1 Functions

The system will perform the following functions:

1. **Create and manage user accounts** to allow users to sign in and maintain personal schedules.
2. **Collect and update weekly availability** through a reusable master schedule or a group-specific schedule.
3. **Record meeting preferences** and distinguish between available, available if necessary, and unavailable time slots.
4. **Create and manage events** with a group name, meeting purpose, duration, candidate dates, and permitted meeting hours.
5. **Generate shareable invitation links** to allow participants to access events and submit availability.
6. **Track availability submissions** and identify missing responses as unknown.
7. **Calculate and rank suitable meeting times** based on participant availability and preferences within the event's scheduling constraints.
8. **Display group availability** through a heatmap that shows participant availability for each time slot.
9. **Confirm and display meeting details** after the organizer selects a final meeting time.
10. **Repeat the scheduling process automatically** at specified intervals for recurring events.

2.2.2 Objectives

- The objectives of the Group Leader application are as follows: we aim to be useful, intuitive, and quick. Firstly, by our objective being useful, we mean that our website should be providing meaning or practical information that was not available before using our service. This means that, for a team of 20, when each member receives a link they can add their schedule, see how many people out of the total have added theirs, and the current best times for the group to meet. With this information, it is useful to them as they do not have to ask everyone their schedule, they do not have to figure out manually what time fits for everyone, and it can be done asynchronously. Thus, this information is of some value and saves time. The second objective we aim to meet is to be intuitive. What being intuitive to us means, is that the website should be easy to use and add their schedule. The website shows lots of useful information such as: how many people have added their schedule thus far, group name and meeting purpose, and the current best time. I also believe that when the group leader creates the schedule link, he can input the possible time the algorithm will select, so for an office group, it will not schedule the meeting at 2am nor display that time. This adds intuitiveness as the algorithm can do more work for the group, and be able to deliver better information to the group. The third objective we will consider is to be quick, where we aim to allow information to be inputted as fast as possible, we can aim to do this through using buttons over text inputs, design choices, and iterating on design to see what UX tester responds to. As well, considering the entire process is key to being quick, such as from when the first person creates the group, to when everyone receives it. For a lot of people, this could be a longer process, so it is important for it to be easy and fast for each person to enter their schedule. Overall, these objectives combine to produce an easy, intuitive, quick, and useful application that provides value to the user through new information they can use in their daily lives.

2.2.3 Constraints

Economic Factors - Our project is free/inexpensive to complete, but must be fully completed by December 3, with a check-in on October 8. This limits our scope by limiting how much time we have to build our project.

Regulatory Compliance (Security & Access) - If Group Leader is to have any identifiable or personal information (such as with accounts), it is a high priority to minimize risks of vulnerabilities in the software.

Reliability - We need to ensure that Group Leader is reliable, and that there will not be problems. It should be able to successfully complete all needed functions without issue.

Sustainability & Environmental Impacts - Group Leader should be easily sustained without any noticeable footprint.

Ethics - The project must not cross any ethical boundaries. A focus on safety/security should solve any ethical problems with the design, but that does not mean other issues cannot arise and we will plan our project to ensure this is not a problem.

Societal Impacts - The project itself is relatively light, when it comes to societal impacts either way. One potential issue is misuse of data the site is given, which we will ensure is not going to happen.

3 Solution

In this section, you will provide an account of some solutions your team brainstormed to implement and test the project. Some solutions might not have all the desired features, some might not satisfy the constraints, or both. These solutions come up in your mind while you brainstorm ways of implementing all the features while meeting the constraints. Towards, the end you select a solution that you think has all the features, testable and satisfies all the constraints. Remember that an engineering design is iterative in nature!

3.1 Solution 1

Write a brief description of your first solution and provide the reasons in terms of testing for not selecting this one.

3.2 Solution 2

This is an improved solution but might not be the final solution that you select. Give a brief description of this solution here. Again focus on its testing attributes.

3.3 Final Solution

This is the final solution. **Explain why it is better than other solutions** (focus more on testing). You may use a table for comparison purposes. After providing the reason for selecting this solution, detail it below.

3.3.1 Components

What components you used in the solution? What is the main purpose of using individual component? What testing method did you employ for each component? Provide a block diagram (with a numbered caption, such as Fig. 1) representing the connectivity and interaction between all the components.

3.3.2 Environmental, Societal, Safety, and Economic Considerations

Explain how your engineering design took into account environmental, societal, economic and other constraints into consideration. It may include how your design has positive contributions to the environment and society? What type of economic decisions you made? How did you make sure that the design is reliable and safe to use?

3.3.3 Test Cases and results

What test suits did you design to test your prototype? How did you execute the test cases to test the prototype?

3.3.4 Limitations

Every product has some limitations, and so is the case with your design product. Highlight some of the limitations of your solution here.

4 Team Work

Since this is a group project, you must have a fair distribution of tasks among yourselves. To this end, you must hold meetings to discuss the distribution of tasks and to keep a track of the project progress.

4.1 Meeting 1

Time: September 17 2026, around 1 pm.

Agenda: Distribution of Project Tasks

Team Member	Previous Task	Completion State	Next Task
James Clinton	N/A	Project Definition, Github creation, Project File Template	2.2.2 Objectives
Peyton Bowen	N/A	Business Case	2.2.1 Functions
Jaydon Forrester	N/A	Cost/Benefit	2.2.3 Constraints

4.2 Meeting 2

Time: Virtual Meeting – Throughout 23 September to 24 September

Agenda: Review of Individual Progress

Team Member	Previous Task	Completion State	Next Task
James Clinton	Project Definition, Github creation, Project File Template	2.2.2 Objectives100%	3.1
Peyton Bowen	Business Case	2.2.1 Functions	3.2

		100%	
Jaydon Forrester	Cost/Benefit	2.2.3 Constraints 100%	Prototyper

4.3 Meeting 3

Provide a similar description here.

4.4 Meeting 4

Provide a similar description here.

5 Project Management

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Any kind of green denotes critical path																
	Task	Slack Time LSD - ESD	Earliest Start Date	Latest Start Date	Start (DD/MM)	End (DD/MM)	Week 1		Week 3		Week 4		Week 6			Week 13	
	First Deliverable																
	Problem Definition	0 Days	17/09/26 1pm	17/09/26 1pm	16/09/26	17/09/26											
	Business Case	0 Days	17/09/26 1pm	17/09/26 1pm	17/09/26	17/09/26											
	Report MQ	0 Days 9 hours	17/09/26 1pm	17/09/26 10pm	17/09/26	17/09/26											
	Project File Template	0 Days 9 hours	17/09/26 1pm	15/09/26 10pm	17/09/26	17/09/26											
	Second Deliverable																
	Design Constraints, Objectives, Functions	5 Days	18/09/26	23/09/26													
	Brainstorm to fulfill objective and constraints	5 Days	18/09/26	23/09/26													
	Third Deliverable																
	Prototyping	0 Days	25/09/26	25/09/26													
	Designing UI	5 Days	25/09/26	30/09/26													
	Programming Project/Backend Dev	5 Days	25/09/26	30/09/26													
	Solution 1 and 2	3 Days	25/09/26	28/09/26													
	Fourth Deliverable																
	Front end																
	Solution 3																
	Final Design Process																

6 Conclusion and Future Work

- A summary of what you achieved. Mention all the design functions and objectives that you achieved while satisfying testing requirements?
- While keeping the limitations of your solution, provide recommendations for future design improvements.

7 References

- Use the IEEE reference style.
- Do not put any reference if it is not cited in the text.

8 Appendix

If you want to provide an additional information, use this appendix.